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FortiBleed – Active Fortinet Credential Harvesting Campaign UPDATE: Indicators of Compromise (IoCs) Available

Introduction

On 17 June 2026, multiple reports emerged regarding an active, large-scale credential harvesting campaign targeting internet-facing Fortinet FortiGate firewalls and SSL VPN appliances – dubbed “FortiBleed”.^{1,2} This came shortly after a security researcher discovered an exposed, unauthenticated operational server containing the threat group's tooling, automation scripts, and a database of verified working credentials.³

FortiBleed is a fully automated, self-perpetuating credential harvesting operation attributed to a Russian-speaking cybercriminal group. The campaign does not exploit any novel zero-day vulnerability; rather, it leverages credentials sourced from prior Fortinet-related data breaches and infostealer malware logs, cycling them at industrial scale against internet-exposed FortiGate management interfaces and SSL VPN endpoints.

The threat actor's advantage derives entirely from credentials never rotated following prior Fortinet-related breach events, enriched with infostealer malware logs harvested from compromised endpoints – reinforcing the cruciality of continuous dark web monitoring for credential exposures.

This is a developing event – our team will continue to monitor for updates and provide further notifications as material developments emerge.

UPDATE: New reporting indicates that in addition to Fortinet, Sophos SSL-VPN, RD-Web, and MSSQL were targeted; further the threat actor behind the FortiBleed campaign is likely operating as an Initial Access Broker.⁴ We issue this follow-up report based on newly available intelligence, including the availability of Indicators of Compromise (IoCs).

Campaign Targeting

Based on public reporting, target selection is assessed to be heavily weighted toward NATO-member country organisations, suggesting a geopolitical dimension alongside financial motives.⁵ Government entities and telecommunications operators are among the most heavily targeted sectors globally.

The operation's scale and targeting breadth — which extends to defence contractors and critical infrastructure operators — suggest motivations beyond purely opportunistic financial crime. We posit

¹ <https://www.hudsonrock.com/blog/fortibleed-75000-fortinet-firewalls-compromised-global-enterprises-exposed-claim-your-ethical-disclosure>

² <https://arcticwolf.com/resources/blog/active-fortibleed-campaign-impacting-fortinet-device-s-across-194-countries/>

³ https://www.linkedin.com/posts/vdyachenko_massive-fortinetfortigate-bruteforceactive-activity-7471222472193830913-YBDi

⁴ <https://socradar.io/wp-content/uploads/2026/06/Dismantling-FortiBleed.pdf>

⁵ <https://www.hudsonrock.com/blog/fortibleed-75000-fortinet-firewalls-compromised-global-enterprises-exposed-claim-your-ethical-disclosure>

that the group operates in a dual capacity: credential monetisation through initial access brokerage, combined with selective targeting of strategic interest to Russian-speaking state-aligned actors.

While the primary targeting weight falls on NATO-affiliated countries, Asian entities have been observed within the affected dataset. Taiwan, Thailand, Malaysia, and Vietnam are specifically identified in researcher reporting. No specific Hong Kong device count has been publicly disclosed. However, Hong Kong is confirmed within the 194 affected countries. Any Hong Kong organisation operating an Internet-accessible FortiGate management interface should treat this as a potential exposure event and refer to the recommendations within this report.

This campaign is also distinct from the Belsen Group leak of January 2025, in which approximately 15,000 static FortiGate configuration files were published on a criminal forum. FortiBleed involves dynamically verified, active credentials — not static historical configurations. The two events are independent; exposure in one does not confirm or exclude exposure in the other.

Attack Mechanics

FortiBleed is a fully automated, three-stage credential harvesting operation.

1. **Automated Scanning for Internet-facing FortiGate devices** on TCP ports 443, 4443, 8443, and 10443. Against each identified device it attempts authentication using a curated credential list derived from prior FortiGate breaches, targeting organisations that never rotated credentials after earlier incidents.
2. **Export Device Configuration and Offline Hash Cracking** – Once authenticated, the attackers export the full device configuration backup. This backup contains significantly more than administrative credentials:
 - Administrator account usernames and hashed passwords
 - SSL-VPN local user credentials
 - Email addresses (only accessible in full config exports)
 - VPN pre-shared keys (PSKs)
 - SNMP community strings

Exported hashes are fed into a dedicated 45-GPU Hashtopolis cracking cluster.⁶ Cracking speed and success rate depends critically on the FortiOS version and whether administrators have re-authenticated since a PBKDF2-capable firmware was applied:

FortiOS Version	Hash Algorithm	Cracking Resistance	Risk Level
Below 7.0.15 / 7.2.11 / 7.4.8	SHA-256 + salt	LOW — crackable in hours to days	CRITICAL
7.2.11, 7.4.8, 7.6.1+ (post admin re-login)	PBKDF2	HIGH — computationally expensive	MEDIUM — verify re-login compliance
7.2.x / 7.4.x (upgraded but no admin re-login)	Old SHA-256 in hidden old-password field	LOW — legacy hash still crackable	HIGH — login-lockout-upon-weaker-encryption must be enabled

⁶ <https://www.bleepingcomputer.com/news/security/fortibleed-leak-exposes-fortinet-vpn-credentials-for-73-000-devices/>

3. **Credential Reuse to Establish Foothold** – Compromised devices subsequently serve as footholds for lateral movement into internal Active Directory environments, enabling credential pivoting and persistence well beyond the initial perimeter breach.

UPDATE: Fortinet has since released an advisory stating that the FortiBleed campaign likely involves the threat actors reusing credentials from previous incidents, such as CVE-2026-24858, CVE-2025-59718, and CVE-2025-59719, along with employing brute-force techniques against devices with weak password hygiene and no multi-factor authentication (MFA).⁷

Detection Opportunities

No traditional network IOCs (C2 IP addresses, malicious domains, or malware file hashes) have been publicly disclosed. FortiBleed operates exclusively via the legitimate FortiGate management interface using stolen credentials. The indicators below are behavioural and log-based.

FortiGate Log Events to Monitor Immediately

Review administrative event logs on all FortiGate devices. CLI: execute log filter category event → execute log filter field subtype admin → execute log display

Log ID	Event	Interpretation
0100032001	Admin login success	Verify every source IP and timestamp. Any unknown source IP should be treated as compromise.
0100032002	Admin login failure	Repeated failures from external IPs = active credential stuffing in progress.
0100044546	Admin account created	Alert immediately — strong indicator of backdoor account creation by attacker.
0100044547	Admin account deleted	Investigate any deletion not initiated by known IT staff.
0100032003	Config backup executed	Any config backup from an unexpected source IP = treat as credential and PSK exfiltration.
Config change events	Firewall policy modification	Compare against last known-good baseline; alert on any unexplained rule additions.

Behavioural Indicators of Attack (Campaign-Level IOAs)

Indicator	Notes
Mass admin login attempts from non-HK / non-APAC IP ranges	Particularly Eastern European and Russian ASNs
Admin logins between 02:00–06:00 HKT from unknown IP addresses	Consistent with attacker operating in European/Russian time zones
Configuration backup executed from IP not matching known management workstations	Stage 2 trigger — treat as confirmed credential and PSK exfiltration
New admin accounts appearing not created by known IT staff	Backdoor persistence mechanism
Firewall policy rules added permitting outbound to unusual cloud storage/endpoints	Potential exfiltration staging or C2 enablement
VPN connections from Alibaba Cloud (AS45102) or Tencent Cloud (AS132203)	Investigate if unexpected in your environment

⁷ <https://www.fortinet.com/blog/psirt-blogs/analysis-of-reported-credential-compromise-of-fortigate-devices>

Safe Reference Versions — FortiOS PBKDF2 Password Hashing

FortiOS Branch	Minimum Safe Version	Notes
7.0.x	7.0.16	Versions below 7.0.15 use SHA-256+salt; upgrade required
7.2.x	7.2.11	Upgrade then require admin re-login to trigger PBKDF2 hashing
7.4.x	7.4.8	Upgrade then require admin re-login to trigger PBKDF2 hashing
7.6.x	7.6.1	Admin re-login required; apply login-lockout-upon-weaker-encryption
8.0.x	8.0.0	PBKDF2 natively enforced

Recommendations**1. Detect and Assess Exposure**

- Immediately check all FortiGate public domains / WAN IP addresses against the Hudson Rock⁸ and SOCRadar⁹ exposure checkers.
- Simultaneously review administrative event logs for the log IDs listed above — particularly 0100032003 (config backup executed) which, if present from an unexpected source IP, indicates Stage 2 exfiltration has already occurred and that VPN PSKs, SNMP community strings, and SSL-VPN user credentials must be treated as compromised.
- Confirm the FortiOS version on all devices against the safe reference versions table; any device below the PBKDF2 threshold is at critical credential cracking risk.

2. Rotate Credentials and Enable PBKDF2 Hardening

- Rotate all FortiGate administrator account passwords immediately on every device, via console or trusted internal management IP — not via the WAN interface. Minimum 16 characters; mix of uppercase, lowercase, numbers, and special characters; never reuse a prior password. Assume all prior passwords are already known to the attacker.
- After rotation, enable the login-lockout-upon-weaker-encryption policy and require all admin accounts to re-authenticate to trigger PBKDF2 re-hashing. Coordinate with all administrators before enabling — accounts that do not re-login will be locked out.
- Additionally rotate all VPN pre-shared keys and SNMP community strings, which are extracted in plaintext during Stage 2 config export and are compromised independently of admin credential cracking:

```
# Rotate admin credentials (repeat for each account)
config system admin
    edit "<admin_account_name>"
        set password <NewStrongPassword_16chars_minimum>
    next
end

# Enable PBKDF2 lockout policy
config system password-policy
    set status enable
    set minimum-length 16
    set must-contain upper-case-letter lower-case-letter number non-
```

⁸ <https://www.hudsonrock.com/fortinet>

⁹ <https://socradar.io/free-tools/fortibleed>

```
alphanumeric
    set login-lockout-upon-weaker-encryption enable
end

# Rotate IPsec VPN PSKs (repeat for each Phase 1 tunnel)
config vpn ipsec phase1-interface
    edit "<tunnel_name>"
        set psksecret <new_strong_psk>
    next
end

# Rotate SNMP community string
config system snmp community
    edit 1
        set name "<new_community_string>"
    next
end
```

3. Block Scanning and Enable Multi-Factor Authentication (MFA)

- Remove HTTPS and SSH administrative access from all public WAN interfaces — management planes must not be reachable from the open internet.
- Restrict all admin accounts to trusted source IP ranges.
- Enable multi-factor authentication on all administrator accounts; FortiToken Mobile is recommended, with email OTP as an interim measure where tokens are unavailable:

```
# Remove HTTPS/SSH admin access from WAN interfaces
config system interface
    edit "<wan_interface_name>"
        unset allowaccess
        set allowaccess ping
    next
end

# Restrict admin accounts to trusted source IPs
config system admin
    edit "<admin_name>"
        set trusthost1 <trusted_mgmt_subnet> <subnet_mask>
    next
end

# Enable FortiToken MFA (recommended)
config system admin
    edit "<admin_name>"
        set two-factor fortitoken
        set fortitoken <token_serial_number>
        set email-to <admin_email@yourcompany.com>
    next
end
```

- Monitor authentication logs continuously for the IOA patterns listed above, particularly login attempts from Eastern European and Russian ASNs and off-hours logins between 02:00–06:00 HKT. Alert immediately on log event 0100044546 (admin account created).

4. Audit Accounts and Treat Confirmed Compromise as Full Incident Response

- Audit all administrator accounts on every FortiGate device and remove any account that cannot be attributed to a known, current staff member or authorised vendor.
- Audit SSL-VPN local user accounts using show user local. If any Stage 2 IOAs are confirmed (e.g., unexpected config backup, unknown admin accounts, or unexplained firewall rule additions) treat the incident as a full network compromise: assume adversary lateral movement into Active Directory has occurred, isolate the device, reset all AD accounts for users whose credentials may have traversed the affected gateway, and engage an incident response provider.
- Do not attempt to remediate in-place on a device where Stage 2 exfiltration is suspected, as the attacker may retain PSK-based VPN access independently of admin password changes.

```
# List all admin accounts
show system admin

# Delete an unknown account
config system admin
    delete "<unknown_account_name>"
end

# List all SSL-VPN local users
show user local

# Delete unknown local user
config user local
    delete "<unknown_user_name>"
end
```

Conclusion

PwC strongly urges organisations leveraging Fortinet FortiGate firewalls and SSLVPN devices perform a threat hunt using the publicly available checker tools. Furthermore, we recommend monitoring for anomalous activity and hardening devices as aforementioned above. **UPDATE:** We strongly advise performing a threat hunt leveraging the newly available Indicators of Compromise (IoCs) captured on subsequent pages.

Indicators of Compromise^{10,11}

IOC Value	Type
hxxp[:]//85[.]11[.]187[.]8:9999	URL
102[.]50[.]247[.]12	IP
103[.]124[.]165[.]105	IP
103[.]154[.]55[.]145	IP
103[.]194[.]242[.]134	IP
103[.]21[.]149[.]215	IP
103[.]50[.]219[.]186	IP
103[.]80[.]60[.]70	IP
103[.]88[.]80[.]163	IP
104[.]160[.]114[.]124	IP
107[.]0[.]184[.]92	IP
146[.]70[.]224[.]23	IP
146[.]70[.]231[.]15	IP
147[.]45[.]45[.]202	IP
152[.]89[.]216[.]207	IP
179[.]43[.]166[.]138	IP
179[.]43[.]166[.]170	IP
185[.]65[.]133[.]193	IP
185[.]65[.]133[.]22	IP
188[.]127[.]226[.]252	IP
188[.]127[.]246[.]183	IP
192[.]253[.]248[.]42	IP
193[.]138[.]7[.]164	IP
193[.]8[.]187[.]10	IP
193[.]8[.]187[.]14	IP
193[.]8[.]187[.]2	IP
193[.]8[.]187[.]22	IP
193[.]8[.]187[.]26	IP
193[.]8[.]187[.]30	IP
193[.]8[.]187[.]34	IP
193[.]8[.]187[.]38	IP
193[.]8[.]187[.]42	IP
193[.]8[.]187[.]46	IP
193[.]8[.]187[.]6	IP
194[.]113[.]39[.]104	IP
194[.]113[.]39[.]105	IP
194[.]113[.]39[.]107	IP
194[.]113[.]39[.]108	IP
194[.]113[.]39[.]109	IP
194[.]113[.]39[.]110	IP

¹⁰ <https://socradar.io/wp-content/uploads/2026/06/Dismantling-FortiBleed.pdf>

¹¹ <https://www.cloudsek.com/blog/inside-the-fortibleed-open-directory-a-technical-analysis-of-what-the-attacker-left-behind>

IOC Value	Type
194[.]113[.]39[.]122	IP
194[.]113[.]39[.]123	IP
194[.]113[.]39[.]125	IP
194[.]113[.]39[.]126	IP
194[.]113[.]39[.]127	IP
194[.]113[.]39[.]131	IP
194[.]113[.]39[.]135	IP
194[.]113[.]39[.]141	IP
194[.]113[.]39[.]142	IP
194[.]113[.]39[.]143	IP
194[.]113[.]39[.]144	IP
194[.]113[.]39[.]145	IP
194[.]113[.]39[.]146	IP
194[.]113[.]39[.]151	IP
194[.]113[.]39[.]158	IP
194[.]113[.]39[.]159	IP
194[.]113[.]39[.]160	IP
194[.]113[.]39[.]161	IP
194[.]113[.]39[.]162	IP
194[.]113[.]39[.]174	IP
194[.]113[.]39[.]176	IP
194[.]113[.]39[.]177	IP
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194[.]113[.]39[.]179	IP
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194[.]113[.]39[.]210	IP
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194[.]113[.]39[.]212	IP
194[.]113[.]39[.]213	IP
194[.]113[.]39[.]214	IP
194[.]113[.]39[.]219	IP
194[.]113[.]39[.]225	IP
194[.]113[.]39[.]226	IP
194[.]113[.]39[.]227	IP

IOC Value	Type
194[.]113[.]39[.]228	IP
194[.]113[.]39[.]229	IP
194[.]113[.]39[.]230	IP
194[.]113[.]39[.]231	IP
194[.]113[.]39[.]238	IP
194[.]113[.]39[.]243	IP
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194[.]113[.]39[.]248	IP
194[.]113[.]39[.]26	IP
194[.]113[.]39[.]45	IP
194[.]113[.]39[.]47	IP
194[.]113[.]39[.]49	IP
194[.]113[.]39[.]53	IP
194[.]113[.]39[.]71	IP
194[.]113[.]39[.]73	IP
194[.]113[.]39[.]75	IP
194[.]113[.]39[.]79	IP
194[.]127[.]167[.]114	IP
213[.]171[.]17[.]74	IP
45[.]144[.]115[.]10	IP
62[.]197[.]149[.]124	IP
77[.]91[.]122[.]11	IP
77[.]91[.]122[.]13	IP
77[.]91[.]122[.]17	IP
77[.]91[.]122[.]31	IP
77[.]91[.]122[.]33	IP
77[.]91[.]122[.]35	IP
77[.]91[.]122[.]39	IP
77[.]91[.]122[.]43	IP
77[.]91[.]122[.]5	IP
77[.]91[.]122[.]7	IP
77[.]91[.]122[.]9	IP
77[.]91[.]96[.]136	IP
80[.]246[.]31[.]105	IP
83[.]48[.]92[.]67	IP
85[.]11[.]187[.]100	IP
85[.]11[.]187[.]101	IP
85[.]11[.]187[.]102	IP
85[.]11[.]187[.]103	IP
85[.]11[.]187[.]105	IP
85[.]11[.]187[.]107	IP

IOC Value	Type
85[.]11[.]187[.]108	IP
85[.]11[.]187[.]109	IP
85[.]11[.]187[.]111	IP
85[.]11[.]187[.]112	IP
85[.]11[.]187[.]113	IP
85[.]11[.]187[.]114	IP
85[.]11[.]187[.]115	IP
85[.]11[.]187[.]116	IP
85[.]11[.]187[.]117	IP
85[.]11[.]187[.]118	IP
85[.]11[.]187[.]119	IP
85[.]11[.]187[.]12	IP
85[.]11[.]187[.]120	IP
85[.]11[.]187[.]121	IP
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85[.]11[.]187[.]157	IP
85[.]11[.]187[.]158	IP
85[.]11[.]187[.]159	IP
85[.]11[.]187[.]16	IP

IOC Value	Type
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85[.]11[.]187[.]162	IP
85[.]11[.]187[.]163	IP
85[.]11[.]187[.]164	IP
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85[.]11[.]187[.]166	IP
85[.]11[.]187[.]167	IP
85[.]11[.]187[.]168	IP
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85[.]11[.]187[.]216	IP
85[.]11[.]187[.]217	IP
85[.]11[.]187[.]218	IP
85[.]11[.]187[.]219	IP
85[.]11[.]187[.]220	IP

IOC Value	Type
85[.]11[.]187[.]221	IP
85[.]11[.]187[.]222	IP
85[.]11[.]187[.]223	IP
85[.]11[.]187[.]227	IP
85[.]11[.]187[.]232	IP
85[.]11[.]187[.]233	IP
85[.]11[.]187[.]234	IP
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85[.]11[.]187[.]68	IP
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85[.]11[.]187[.]74	IP
85[.]11[.]187[.]76	IP
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85[.]11[.]187[.]90	IP
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85[.]11[.]187[.]94	IP
85[.]11[.]187[.]98	IP
87[.]249[.]133[.]20	IP
87[.]249[.]133[.]79	IP
89[.]187[.]163[.]211	IP
91[.]214[.]78[.]143	IP
95[.]214[.]217[.]25	IP
96[.]18[.]56[.]78	IP
96[.]18[.]57[.]90	IP
96[.]43[.]51[.]7	IP
99[.]9[.]111[.]82	IP
0fd72c5f46b5718beef800735d9e0f27f53fbf7be6bde2805cff810e87152850	SHA256
1c3e93a6b447603279460eed5104dceb708fa2ed2d623541bc69c0dfb869929e	SHA256
2b66e742dc492daaaf5e112e12e2b057c216e0a89292773eb37799ecf8626323	SHA256

IOC Value	Type
2c98c86e6bd6f46cbd6c89d855541b9da91515b1bb986641a77e31c5c6aa2abb	SHA256
2F0329B0DE0B3A1DC1D866665AE5339C6B333E6DB566F3EDE1332A55C54168CC	SHA256
43f21f040c3006498ef3fd72554f083080a47cf9c563d7cdfba34a8d60c19b44	SHA256
45917803009456a0c703fa9a8fadac8f2a6e2499e89a6be1c8198665381a6fce	SHA256
4618112eecf25a9ed970145305eff0839365f2998f23f8cdedcd3e2224051d3d	SHA256
4d0b62d3162d4be391e3ba1e191dad28e5e5d5b161cfdef60eeb4361a92d8413	SHA256
80d83eb01f28c87a61b51f1f83805e63a791905f019bd3b87f10a10f66efab1e	SHA256
810f7e72fffb9c7793f789e9a3510b4550a02c8f69d24ac3acad8fb715d20c68d	SHA256
9a3cddb8e36063ab639f1532a9d6e73ffc9511e18fcc541ce76cbcbca3f19846ec	SHA256
a474e04340a6425914b110bcd55da50a5f3f618f8b36cb4da4bfa6bf1d3804b4	SHA256
a8b09fd4f7ff2f298b45ca602992f44b3c2ac3746bcd182c59ab2a20c690954	SHA256
b76d83918473be1550db8fe7bf60479841599bc5b0c30e2d1184432b99c7ff02	SHA256
c1406935ebcf840acca58b2da12ddedae5bc029b8658ff12272962d98e28d495	SHA256
c90adcfb6ac7ae5e250d52b2f09e0418375fc8af36f19642f06c5efde765c86b	SHA256
d0ab429f1289fc91dfcaef28ccc768c61f6662ee15d7eaf9ca3d0dee67b1e61	SHA256
d4f39302a3a2a99fddd5da91e02d89f5023e166ac99939576b5d78ef76103899	SHA256
dd781aab0b946d610fd1ec3a3b8e53f8da9800b0bd8c407dcf69a089e78e0f3e	SHA256
e056ad0fece00c400f3439ed9810d5be5c00ae63a852e4be2bb9ccfd6ac77766	SHA256
ea914c06556b3319ee0d2dc116fec2fd3ef836e3a956dcb538d753c95bd6cf71	SHA256
f16bc1fac8b8b951d35d6c94e8c7257f5005e8963bf075d079964269d0b931f7	SHA256
f841ebe0f47f332704f038256fa7b7f79aa910c964c81d1d8513cc9b36b77630	SHA256
fa36ad03e92e0399ec4eea7ba37e4a35fd7dc4391558f8f6e9899bce93095f5d	SHA256
85[.]11[.]187[.]8:8443	IP:port
85[.]11[.]187[.]28	IP
185[.]229[.]26[.]83	IP
213[.]169[.]49[.]142	IP
38[.]117[.]87[.]37	IP
198[.]53[.]64[.]194	IP
175[.]155[.]64[.]221	IP

Further information

If you need any further advice or would like PwC's leading global incident response team to support you, please do not hesitate to contact us.

This report has been provided to clients as part of PwC's Threat Intelligence Team Cyber-as-a-Service offering. More detailed analysis on the topics covered in this report can be provided on request.

If you would like more information on any of the threats discussed in this alert please feel free to get in touch, by emailing darklab.cti@hk.pwc.com.

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